SYED AMMAL ARTS AND SCIENCE COLLEGE

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Department of Information Technology

NAAN MUDHALVAN PROJECT DOCUMENTATION

CLASS: III B.SC., IT

SEMESTER: VI

SUBJECT CODE: 25NMAAA

SUBJECT NAME: AGENTIC AI APPLICATION DEVELOPER PROJECT

TITLE: HTML QUICK-STYLER

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HTML QUICK STYLER

AI-Powered HTML Quick Styler using Intelligent Layout and Design Automation

PROJECT OVERVIEW

- HTML Quick Styler is an AI-powered web page generation system designed to automatically create structured, responsive, and visually appealing HTML web pages based on user input.
- The system uses the IBM Granite 3.3 2B Instruct model to understand the user's description and generate complete HTML and CSS code.
- The application eliminates the need for manual coding by automating layout creation, styling, and structure building. It provides a live preview feature and allows users to download production-ready HTML code.
- This project demonstrates how Artificial Intelligence can simplify modern web development by reducing development time and improving design consistency.

PROBLEM STATEMENT

- Designing professional and responsive websites requires strong knowledge of HTML and CSS.
- Many beginners struggle with layout structuring, alignment, and responsive design implementation.
- Manual coding of common sections like header, hero, and footer is repetitive and time-consuming.
- Maintaining consistent color themes and modern UI standards is often challenging.
- Existing website builders limit customization or do not provide clean structured code.

- HTML Quick Styler solves this problem by:
- Automatically generating structured semantic HTML.
- Applying responsive and modern CSS styling.
- Providing professional predefined color themes.
- Reducing manual coding effort and development time.
- Delivering clean, production-ready downloadable code.
- Ensuring consistency and improved user experience across devices

CORE TECHNOLOGIES USED

- Python 3.8+
- IBM Granite 3.3 2B Instruct Model
- Gradio – Interactive Web Interface
- HTML5 – Web Page Structure
- CSS3 – Styling and Responsive Design
- JavaScript – Basic Interactivity
- Google Colab – Development & Deployment Environment

SYSTEM ARCHITECTURE

The project consists of four major modules:

1. HTML GENERATION ENGINE

• Extracts page title and user description

• Detects required sections such as hero, about, features, and contact

• Generates structured semantic HTML layout

2. CSS STYLING ENGINE

• Applies modern UI styling and layout structure

• Implements responsive design using Flexbox and Grid

• Ensures proper spacing, alignment, and typography

3. USER INTERFACE (GRADIO)

• Title input area
• Description input area
• Theme selector (color styles)
• Output display with generated HTML code

4. EXPORT MODULE

• Download generated HTML file
• Copy production-ready HTML code
• Enable easy sharing and reuse of generated pages

PROJECT WORKFLOW

Phase 1:

• Setup & Environment Configuration
• Install required dependencies
• Configure IBM Granite model
• Setup development environment in Google Colab

Phase 2:

• HTML Generation Development
• Extract page title and description
• Detect required web page sections
• Generate structured semantic HTML layout

Phase 3:

• CSS Styling Implementation
• Apply responsive layout using Flexbox and Grid
• Implement modern UI styling
• Add predefined color themes

Phase 4:

• Interface & Export
• Build interactive Gradio UI
• Implement live preview functionality
• Enable HTML code download

Phase 5:

• Testing & Deployment
• Functional testing with sample inputs
• Validate responsive design
• Deployment using Google Colab public URL

KEY FEATURES

- Automatic HTML web page generation
- Responsive layout with modern CSS styling
- Multiple professional color themes
- Live preview of generated web pages
- Downloadable production-ready HTML code
- User-friendly web interface

APPLICATIONS

- Portfolio website creation
- Business and startup landing pages
- Educational web development projects
- Rapid website prototyping
- Learning tool for beginner developers
- Front-end design automation

CODING

Phase 1

```
!pip install gradio transformers accelerate torch
```

Phase 2

```
import gradio as gr

from transformers import AutoTokenizer, AutoModelForCausalLM

import torch
```

Phase 3

```
model_name = "ibm-granite/granite-3.3-2b-instruct"

tokenizer = AutoTokenizer.from_pretrained(model_name)

model = AutoModelForCausalLM.from_pretrained(

    model_name,

    torch_dtype=torch.float16,

    device_map="auto"

)
```

Phase 4

```
import gradio as gr

PALETTES = {

    "modern": {

        "Primary": "#667eea",

        "Secondary": "#764ba2",

        "Accent": "#f97316",

        "Background": "#0f172a",

        "Text": "#f1f5f9"

    }

  "ocean": {

        "Primary": "#2193b0",

        "Secondary": "#6dd5ed",

        "Accent": "#00bcd4",

        "Background": "#0c1e2c",

        "Text": "#e0f2fe"

    },
```

"forest": {
    "Primary": "#134e4a",
    "Secondary": "#16a34a",
    "Accent": "#22c55e",
    "Background": "#0f1f17",
    "Text": "#dcfce7"
},
"vibrant": {
    "Primary": "#ff416c",
    "Secondary": "#ff4b2b",
    "Accent": "#ffe600",
    "Background": "#1a1a2e",
    "Text": "#ffffff"
},
"sunset": {
    "Primary": "#ff7e5f",
    "Secondary": "#feb47b",

"Accent": "#ffcc70",
"Background": "#2c1b18",
"Text": "#fff3e6"
},
"luxury": {
"Primary": "#b993d6",
"Secondary": "#8ca6db",
"Accent": "#d4af37",
"Background": "#0d0d0d",
"Text": "#f5f5f5"
},
"creative": {
"Primary": "#00c6ff",
"Secondary": "#0072ff",
"Accent": "#ff00cc",
"Background": "#141e30",
"Text": "#e0e0ff"

```
    }
}
```

Phase 5

```
def generate_page(title, description, style):
    palette = PALETTES.get(style, PALETTES["modern"])
    return f"""
<!DOCTYPE html>
<html>
<head>
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<style>
:root {{
    --primary:{palette["Primary"]};
    --secondary:{palette["Secondary"]};
    --accent:{palette["Accent"]};
    --bg:{palette["Background"]};
```

```
    --text:{palette["Text"]};
}}
body {{
    margin:0;
    font-family:Arial, sans-serif;
    background:var(--bg);
    color:var(--text);
    scroll-behavior: smooth;
}}
.navbar
    {{ background:#0b11
    20; padding:20px
    60px; display:flex;
    justify-content:space-between;
}}
.navbar a
    {{ color:var(--
    text);
```

```
    text-decoration:none;
    margin-left:30px;
}}
.hero {{
    height:80vh;
    background:linear-gradient(135deg,var(--primary),var(--
secondary));
    display:flex;
    flex-direction:column;
    justify-content:center;
    align-items:center;
    text-align:center;
}}
.hero h1 {{
    font-size:48px;
    margin-bottom:20px;
}}
```

```
.hero p {{
    font-size:18px;
}}
.hero button
    {{ margin-top:20px;
    padding:14px 35px;
    border:none;
    border-radius:8px;
    background:var(--accent);
    color:white;
    font-weight:bold;
    cursor:pointer;
}}
section
    {{ padding:100px
    20px; text-
    align:center;
}}
```

```
footer
    {{ background:#0b11
    20; padding:40px;
    text-align:center;
}}
</style>
</head>
"""
```

Phase 6

```
dark_css = """
body { background:#0b1120 !important; }
.gradio-container
    { background:#0b1120 !importa
    nt; max-
    width:1400px !important;
    margin:auto !important;
}
```

```
h1, h2, h3, p, label
    { color:#f1f5f9 !important;
}
textarea, input, select
    { background:#111827 !important;
    color:white !important;
    border:1px solid #2d3748 !important;
    border-radius:8px !important;
}
button {
    background:#f97316 !important;
    color:white !important;
    border:none !important;
    border-radius:8px !important;
    font-weight:600 !important;
    padding:12px 20px !important;
    cursor:pointer;
```

```
}
button:hover
    { opacity:0.9;
}
"""
```

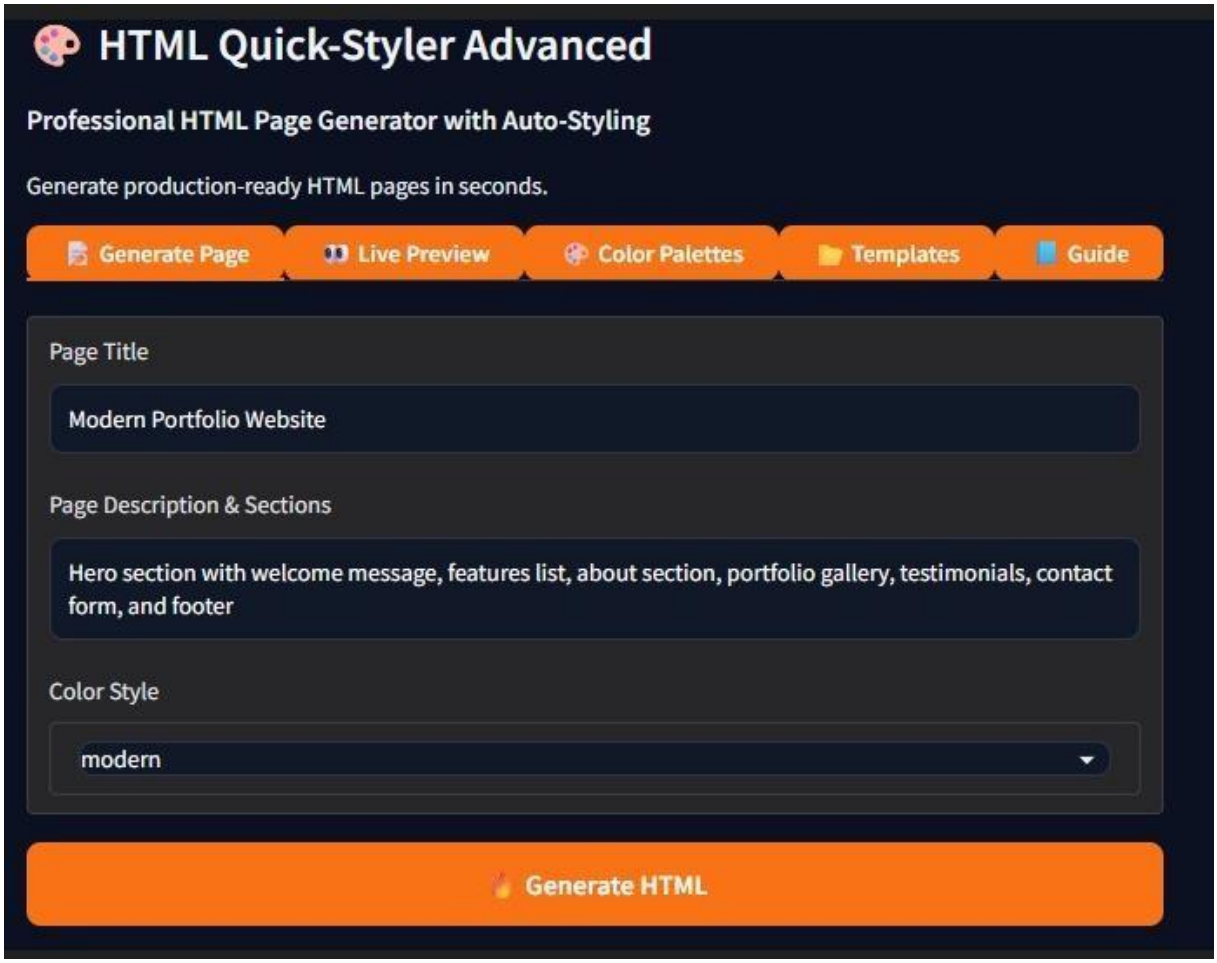
Phase 7

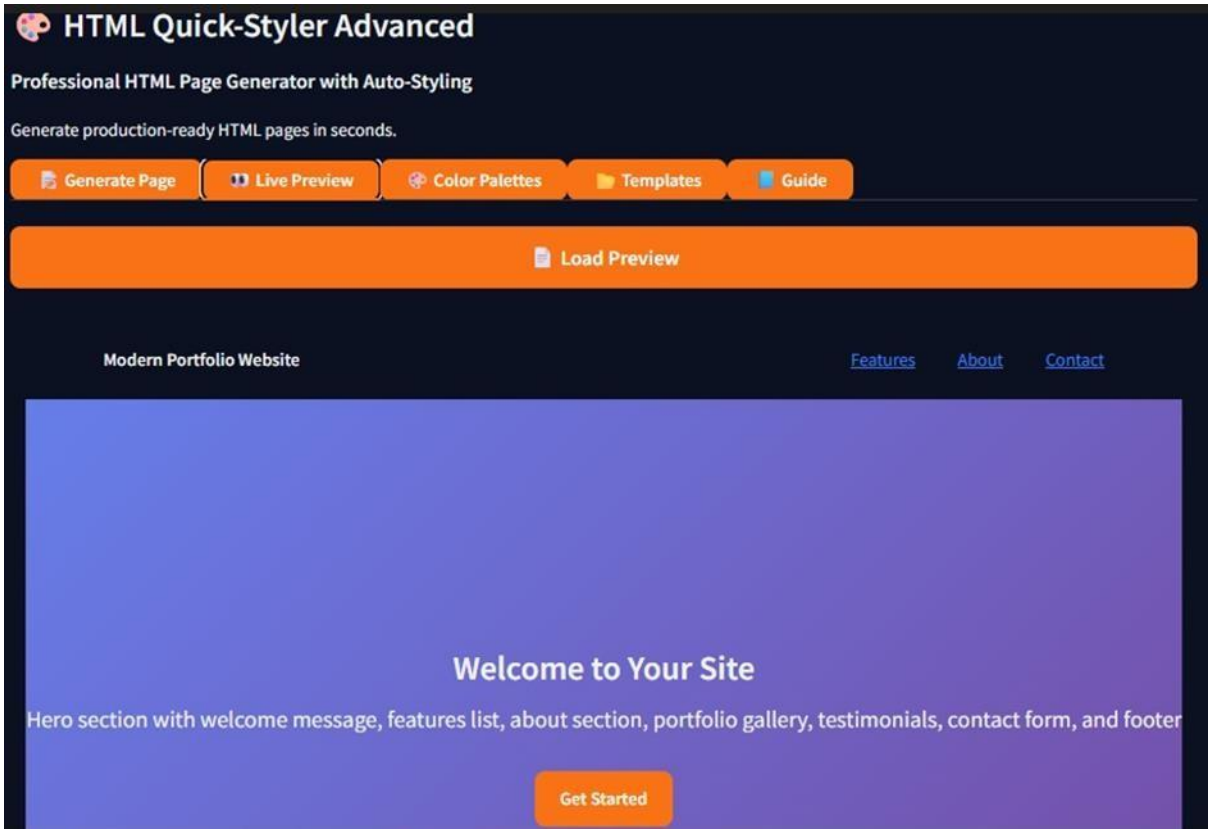
```
with gr.Blocks(css=dark_css) as demo:
    gr.Markdown("""
#     HTML Quick-Styler Advanced
### Professional HTML Page Generator with Auto-Styling
Generate production-ready HTML pages in seconds.
""")
```

Phase 8

```
demo.launch()
```

OUTPUT:


HTML Quick-Styler Advanced
Professional HTML Page Generator with Auto-Styling
Generate production-ready HTML pages in seconds.
Generate Page
Live Preview
Color Palettes
Templates
Guide
Page Title
Modern Portfolio Website
Page Description & Sections
Hero section with welcome message, features list, about section, portfolio gallery, testimonials, contact form, and footer
Color Style
modern
Generate HTML


HTML Quick-Styler Advanced
Professional HTML Page Generator with Auto-Styling
Generate production-ready HTML pages in seconds.
Generate Page
Live Preview
Color Palettes
Templates
Guide
Load Preview
Modern Portfolio Website
Features
About
Contact
Welcome to Your Site
Hero section with welcome message, features list, about section, portfolio gallery, testimonials, contact form, and footer
Get Started

HTML Quick-Styler Advanced

Professional HTML Page Generator with Auto-Styling

Generate production-ready HTML pages in seconds.

Generate Page | Live Preview | Color Palettes | Templates | Guide

Choose Style

modern | ocean | forest | vibrant | sunset | luxury | creative

Get Palette

Primary: #667eea
Secondary: #764ba2
Accent: #f97316
Background: #0f172a
Text: #f1f5f9

Use via API · Built with Gradio · Settings

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Quick Templates

Portfolio Website

Hero, Portfolio Gallery, About, Contact, Footer

Use Template

SaaS Landing Page

Hero, Features, Pricing, Testimonials, CTA, Footer

Use Template

Company Website

Header, About, Services, Team, Contact, Footer

Use Template

Use via API · Built with Gradio · Settings

🎨 HTML Quick-Styler Advanced

Professional HTML Page Generator with Auto-Styling

Generate production-ready HTML pages in seconds.

📄 Generate Page | 👁 Live Preview | 🎨 Color Palettes | 📁 Templates | 📘 Guide

📖 HTML Quick-Styler Guide

How it works

1 Enter Details

- Title: Page name
- Description: Section you want
- Color Style: pick a theme

2 Automatic Detection

The tool recognizes these keywords:

- hero → Hero section
- features → Features Grid
- about → About Section
- portfolio → Portfolio
- testimonial → Testimonial
- pricing → Pricing
- contact → Contact

CONCLUSION

- **HTML Quick Styler** successfully demonstrates how automated design generation combined with a modern web interface can simplify the process of creating professional HTML web pages.

- It helps users quickly generate structured, responsive, and visually appealing web layouts without requiring advanced front-end development knowledge.

- The system provides an efficient and user-friendly solution for rapid website prototyping, improving productivity and making web design more accessible for developers, students, and beginners